

STD asymptote analysis: why b_E hits a 0.2 floor

train-srnn

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Setup

Consider a single excitatory neuron in the `srnn-std-e-only` preset (`train_srnn/models/srnn_cell.py:135`) with all initial-condition parameter values held at their defaults. The relevant cell-internal flags for this preset are

$$n_{a,E} = 0, \quad n_{a,I} = 0, \quad n_{b,E} = 1, \quad n_{b,I} = 0,$$

so spike-frequency adaptation (SFA) is disabled and only short-term synaptic depression (STD) is active, on the excitatory side. We isolate a single neuron driven by a constant external input u , taking the recurrent contribution $W_{\text{eff}}(b \cdot r)$ to be zero (or absorbed into u) for the cleanest closed-form analysis. We address the recurrent contribution at the end.

The dendritic state of the neuron is $x(t)$, the synaptic-depression state is $b(t) \in [0, 1]$ (we drop the E subscript for the rest of the analysis), the instantaneous firing rate is $r(t) \in [0, 1]$, and the firing threshold is a_0 .

The key default parameter values, from `srnn_cell.py` initialization (lines 215, 224, 278–279), are

$\tau_d = 0.10$ s	(dendritic membrane time constant)
$\tau_{\text{rec}} = 1.00$ s	(STD recovery)
$\tau_{\text{rel}} = 0.25$ s	(STD release / depression)
$a_0 = 0.35$	(firing threshold).

The activation function is the piecewise sigmoid $\sigma(\cdot)$ defined in `srnn_cell.py:40--75`, with output bounded in $[0, 1]$ and saturating at 1 for input > 0.55 .

1. Dendritic dynamics

From `_compute_rhs` (`srnn_cell.py:484--485`), the dendritic ODE in the single-neuron case (no recurrence) is

$$\tau_d \frac{dx}{dt} = -x + u + W_{\text{eff}}(b \cdot r). \quad (1)$$

With identity input drive of constant amplitude u and zero recurrence, this reduces to

$$\tau_d \frac{dx}{dt} = -x + u, \quad (2)$$

which is a first-order linear ODE with solution

$$x(t) = u + (x(0) - u)e^{-t/\tau_d}. \quad (3)$$

For $x(0) \approx 0$ (the cell initializes the dendrite at $0.1 \cdot \mathcal{N}(0, 1)$, so $|x(0)|$ is small), the dendrite settles to $x_\infty = u$ on a timescale of order $\tau_d = 0.1$ s. After roughly $5\tau_d = 0.5$ s, $x \approx u$ to within 1%.

2. Firing rate

The firing rate is computed in `srnn_cell.py:470` as

$$r(t) = \sigma(x(t) - a_0). \quad (4)$$

The piecewise sigmoid σ saturates at $r = 1$ once its argument exceeds 0.55. So firing saturates at $r = 1$ once

$$x(t) - a_0 > 0.55, \quad \text{i.e.} \quad x(t) > 0.90. \quad (5)$$

For a step input u with $u > 0.9$, the dendrite settles to $x \approx u > 0.9$ after the transient, and r then saturates at $r = 1$ indefinitely.

The crucial observation is that r is bounded above by 1. No matter how strong the input, the firing rate cannot exceed unity, because the activation function caps it. This bound is what produces the floor on b derived below.

3. STD dynamics

From `_compute_rhs` (`srnn_cell.py:505`), the STD ODE for the excitatory neuron is

$$\frac{db}{dt} = \frac{1-b}{\tau_{\text{rec}}} - \frac{r \cdot b}{\tau_{\text{rel}}}. \quad (6)$$

Two opposing terms:

- the recovery term $(1-b)/\tau_{\text{rec}}$ pulls b back toward 1 (no depression);
- the release term $r \cdot b/\tau_{\text{rel}}$ depresses b at a rate proportional to both the current firing rate r and the current depression level b .

At $b = 1$ (rested) the recovery is zero and depression dominates as soon as $r > 0$. At $b = 0$ the depression is zero and recovery dominates. The dynamics are therefore a competition between a linear restoring force and a multiplicative depletion.

4. Steady-state derivation

Assume the dendrite has settled and r is constant ($r = r_\infty$). Setting $db/dt = 0$ in equation~(6) gives

$$0 = \frac{1-b_\infty}{\tau_{\text{rec}}} - \frac{r_\infty \cdot b_\infty}{\tau_{\text{rel}}}, \quad (7)$$

$$\frac{1-b_\infty}{\tau_{\text{rec}}} = \frac{r_\infty \cdot b_\infty}{\tau_{\text{rel}}}, \quad (8)$$

$$\tau_{\text{rel}}(1-b_\infty) = \tau_{\text{rec}} r_\infty b_\infty, \quad (9)$$

$$\tau_{\text{rel}} = b_\infty (\tau_{\text{rec}} r_\infty + \tau_{\text{rel}}), \quad (10)$$

$$\boxed{b_\infty(r_\infty) = \frac{\tau_{\text{rel}}}{\tau_{\text{rec}} r_\infty + \tau_{\text{rel}}}.} \quad (11)$$

This is the Tsodyks–Markram steady-state expression; equation~(11) is monotonically decreasing in r_∞ .

5. Numerical asymptote at the IC parameter values

Plugging the default values $\tau_{\text{rec}} = 1.0$ s and $\tau_{\text{rel}} = 0.25$ s into equation~(11) as a function of r_∞ :

$$b_\infty(r_\infty) = \frac{0.25}{1.0 \cdot r_\infty + 0.25} = \frac{1}{4r_\infty + 1}.$$

Selected values:

r_∞	b_∞
0.0	1.000
0.1	0.714
0.25	0.500
0.5	0.333
0.75	0.250
1.0	0.200

Because $r_\infty \in [0, 1]$ (bounded by the saturating piecewise sigmoid in section 2), the minimum of $b_\infty(r_\infty)$ over the allowed range is achieved at $r_\infty = 1$ and equals

$$\boxed{b_\infty^{\min} = \frac{\tau_{\text{rel}}}{\tau_{\text{rec}} + \tau_{\text{rel}}} = \frac{0.25}{1.25} = 0.20.} \quad (12)$$

This is the asymptotic floor: a single neuron driven hard enough to saturate its firing rate cannot depress its synaptic-depression state b below 0.20 at the IC parameter values, no matter how long the input persists.

The threshold $b_\infty = 0.5$ is hit at the modest firing rate $r_\infty = \tau_{\text{rel}}/\tau_{\text{rec}} = 0.25$. Any neuron firing at $r > 0.25$ will sustain $b < 0.5$ at steady state. So crossing the 0.5 line in numerical experiments is easy; the 0.2 floor is the hard limit set by full saturation.

6. Time constant of approach

For constant r_∞ , equation~(6) is linear in b :

$$\frac{db}{dt} = \frac{1}{\tau_{\text{rec}}} - b \left(\frac{1}{\tau_{\text{rec}}} + \frac{r_\infty}{\tau_{\text{rel}}} \right). \quad (13)$$

This has the form $\dot{b} = -(b - b_\infty)/\tau_{\text{eff}}$, with

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_{\text{rec}}} + \frac{r_\infty}{\tau_{\text{rel}}}, \quad \tau_{\text{eff}}(r_\infty) = \frac{\tau_{\text{rec}} \tau_{\text{rel}}}{\tau_{\text{rel}} + r_\infty \tau_{\text{rec}}}. \quad (14)$$

At the IC parameter values:

r_∞	τ_{eff} (s)
0.0	1.000
0.25	0.500
0.5	0.333
1.0	0.200

At full saturation, $\tau_{\text{eff}} = 0.2$ s, so depression is essentially complete after $5\tau_{\text{eff}} = 1.0$ s of sustained firing. This matches the empirical observation in `scripts/diag_std_b_e_step.py` that b reaches the 0.2 asymptote within the first second of a strong step.

The closed-form trajectory for a step input applied at $t = 0$, with $b(0) = 1$, is

$$b(t) = b_\infty + (1 - b_\infty) e^{-t/\tau_{\text{eff}}}. \quad (15)$$

7. Why $r_\infty < 1$ is the typical case

The floor $b_\infty^{\text{min}} = 0.2$ is reached only when the neuron is driven hard enough to saturate its firing rate. In a realistic network this is often not the case:

- Recurrent inhibition (Dale’s law gives an explicit I population) suppresses E firing, so a typical E neuron in the trained network operates at r well below 1.
- For zero-mean inputs (as with z-scored SEEG channels), driving x above the saturation threshold $x > 0.9$ requires either large positive excursions in the input or strong recurrent excitation.
- Once trained, a_0 may have shifted away from its initial value 0.35, raising or lowering the effective threshold for saturation.

At the modest typical firing rates achieved during SEEG training (empirically order $r \approx 0.1$ – 0.3), the steady-state b from equation~(11) is in the range 0.45–0.71, all above 0.5. This is consistent with the observation that trained models’ b_E rarely depresses below 0.5 in practice, despite the cell correctly producing a 0.2 asymptote when given a strong artificial step (see the diagnostic in `scripts/diag_std_b_e_step.py`).

8. Effect of recurrent contributions

Restoring the recurrent term in the dendrite ODE gives

$$\tau_d \frac{dx}{dt} = -x + u + W_{\text{eff}}(b \cdot r). \quad (16)$$

For a single neuron embedded in the network, $W_{\text{eff}}(b \cdot r)$ contributes an additional drive that can either push x above the saturation threshold (recurrent excitation) or pull it down (recurrent inhibition via Dale’s law). The qualitative analysis is unchanged:

- r remains bounded in $[0, 1]$ since it is the output of σ ;

- the steady-state $b_\infty(r)$ from equation (11) still applies, with r taking the value induced by the combined external + recurrent drive at steady state;
- the floor $b_\infty^{\min} = 0.2$ from equation (12) is unchanged.

Recurrence shifts where on the curve $b_\infty(r_\infty)$ a given neuron operates, but does not lower the floor.

9. Lower-bound interpretation

The 0.2 floor has a clean physical interpretation: $\tau_{\text{rel}}/(\tau_{\text{rec}} + \tau_{\text{rel}})$ is the fraction of synaptic resources that, at maximum sustained activity, are in a usable (not depressed) state. With the IC values, 1/5 of resources remain available even when the neuron is firing at maximum rate, because the recovery process operates at a finite rate ($1/\tau_{\text{rec}} = 1$ Hz at default) regardless of how strong the release is.

To enable b to fall below 0.2, one of the following would have to change:

- Lower τ_{rel} relative to τ_{rec} (faster release): e.g. $\tau_{\text{rel}} = 0.05$ would give floor $0.05/1.05 \approx 0.048$.
- Increase τ_{rec} relative to τ_{rel} (slower recovery): e.g. $\tau_{\text{rec}} = 4$ would give floor $0.25/4.25 \approx 0.059$.
- Allow r to exceed 1 (e.g. by replacing the bounded sigmoid with an unbounded activation), in which case $b_\infty = \tau_{\text{rel}}/(\tau_{\text{rec}} \cdot r + \tau_{\text{rel}})$ would continue decreasing past 0.2.

None of these are happening at IC parameter values; hence the floor.

10. Summary

- The STD ODE in the cell is the standard Tsodyks–Markram form $\dot{b} = (1 - b)/\tau_{\text{rec}} - r b/\tau_{\text{rel}}$.
- Steady-state: $b_\infty(r) = \tau_{\text{rel}}/(\tau_{\text{rec}} r + \tau_{\text{rel}})$.
- At IC values $\tau_{\text{rec}} = 1.0$, $\tau_{\text{rel}} = 0.25$, $b_\infty(1) = 0.20$ is the minimum.
- Approach time constant at $r = 1$: $\tau_{\text{eff}} = 0.20$ s. Full depression in ~ 1 s.
- The bounded firing rate $r \in [0, 1]$ from the piecewise sigmoid caps depression at this floor.
- Trained models that operate at $r \ll 1$ on average never reach this floor, even though the underlying ODE permits it (this is what the diagnostic script confirms numerically).